INCH POUND
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MILITARY SPECIFICATION

SCREW THREADS, INTERNAL COLD FORMED GENERAL SPECIFICATION FOR

THIS SPECIFICATION IS APPROVED FOR USE WITHIN THE US ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND, DEPARTMENT OF THE ARMY, AND IS AVAILABLE FOR USE BY ALL DEPARTMENTS AND AGENCIES OF THE DEPARTMENT OF DEFENSE.

1. SCOPE

1.1 Scope. This specification covers screw threads, internal cold formed and the acceptance practice used to determine conformance to the requirements.

*1.2 Classification. Screw threads shall be Unified form and series in accordance with FED-STD-H28/2 and shall be of the following classes:

Class 2B
Class 3B

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications and standards. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

STANDARDS

FEDERAL

FED-STD-H28/2 - Unified Inch Screw Threads - UN and UNR Thread Forms.

FED-STD-H28/6 - Gages and Gaging for Unified Screw Threads - UN and UNR Thread Forms.

FED-STD-H28/20- Inspection Methods for Acceptability of UN, UNR, UNJ, M, and MJ Screw Threads.

MILITARY

MIL-STD-1949 - Inspection, Magnetic Particle.

MIL-STD-6866 - Inspection, Liquid Penetrant

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless exemption has been obtained.

3. REQUIREMENTS

*3.1 Thread configuration (see 6.3). The form of thread configuration, dimensions, thread classes, and thread series shall be in accordance with FED-STD-H28/2.

3.1.1 Gaging system. Gaging system for acceptability shall be selected and specified in accordance with FED-STD-H28/20.

*3.1.2 Coated thread allowance. Unless otherwise specified, when internally threaded parts are to be coated, the threads may be overcut to allow for the coating thickness in accordance with FED-STD-H28/2.

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NOTE: If threads are not overcut, allowance for coating will reduce the pitch diameter tolerance available for the tapping process. Modified limits must be considered in the design of cold forming taps for coated threads.

3.1.3 Thread allowance for dry film lubricant. Internal threads to be coated with dry film lubricant may have the maximum pitch diameter increased by not more than 0.001 inch. The increase in pitch diameter to accommodate dry film lubricant is not in addition to that specified if another coating is applied before the dry film lubrication. The dry film lubrication may be removed for gaging. If the parts are to be returned to stock, they shall be recoated with dry film lubricant.

3.1.4 Lead and flank angle. Equivalent change in functional diameter due to variations in lead (including helix) or flank angle shall not exceed one-half the total pitch diameter tolerance for each element individually. These maximum variations and the pitch diameter tolerance cannot be applied simultaneously.

*3.1.5 Circularity (roundness) (see 6.3.2). Screw thread pitch diameter shall be circular within a tolerance zone of one-half the pitch diameter tolerance where pitch diameter tolerance is less than 0.004 inch. Where pitch diameter tolerance is 0.004 inch or larger tolerance zone is 0.002 inch. When circularity is checked using pitch diameter indicating gage segments or rolls, circularity is equal to one-half the difference between maximum and minimum pitch diameter readings. Threads 1.5000 inches and larger with 16 threads per inch or less may exceed the tolerance by 0.002 inch over a maximum arc of 15 degrees, in the direction of minimum material (oversize thread) in this area, provided that this overcut does not result in raised material on the thread flanks or roots.

3.1.6 Taper. Taper of the pitch diameter over the length of engagement shall be within but not exceed one-half the pitch diameter tolerance. Unless otherwise specified, length of engagement used shall be considered equal to the basic major diameter for the UNC, UNF, and 8 UN series and 9 pitches for all other thread series.

*3.1.7 Runout. The circular runout of the minor diameter cylinder with the pitch diameter cylinder shall not exceed twice the pitch diameter tolerance (see 6.3.1).

3.2 Grain flow. The grain flow in the threads shall be continuous and shall follow the general contour with the maximum density at the bottom of the root as shown in figure 1 in conformance with 4.3.

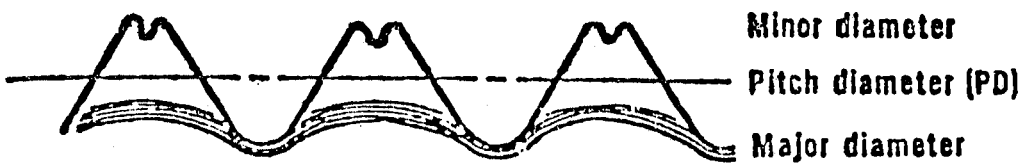


FIGURE 1. Thread grain flow.

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*3.3 Discontinuities in threads. Permissible and nonpermissible thread defects are defined in 3.3.1 and 3.3.2.

*3.3.1 Root and flank defects. Root and flank defects such as laps, seams, notches, slivers, folds, roughness, and oxide scale are not permissible (see figure 2).

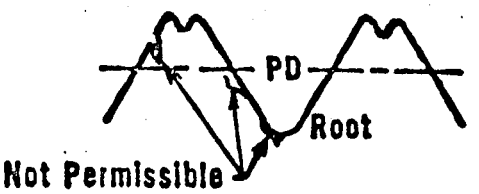
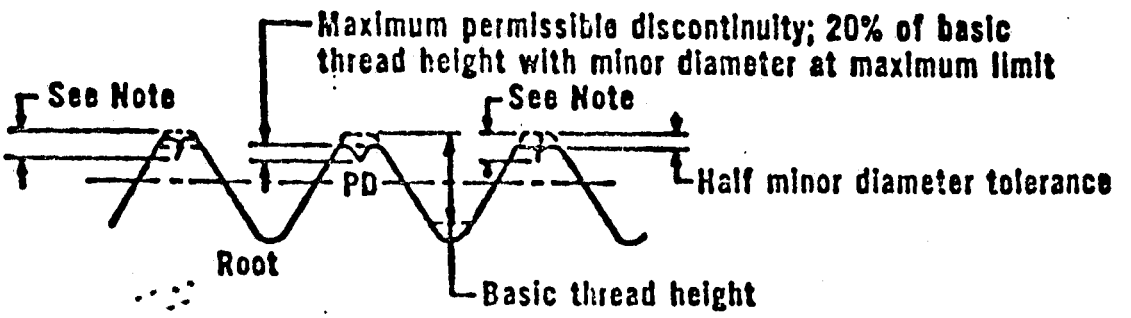


FIGURE 2. Root and flank defects.

*3.3.2 Crest imperfections. Crest craters, crest laps, or a crest lap in combination with a crest crater are permissible provided that the imperfections do not extend deeper than 20% of the basic thread height (see Table 1) as measured from the thread crest when the thread minor diameter is at maximum size (see figure 3). The minor diameter of the thread shall be measured prior to sectioning. As the minor diameter of the thread approaches minimum size, values for depth of crest crater and crest lap imperfections listed in table I may be increased by one-half of the difference between the maximum minor diameter and the actual minor diameter as measured on the part.



NOTE: Depth of discontinuity equals 20 percent of basic thread height plus half the difference between actual minor diameter and maximum minor diameter.

FIGURE 3. CREST IMPERFECTIONS.

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4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

*4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Quality conformance inspection. Quality conformance inspection applicable to screw threads shall be governed by the quality assurance provisions, including sampling plans, of the pertinent product procurement document and 4.3, 4.4, and 4.5 herein.

4.3 Thread form and dimensions. Product thread form and dimensions shall be accepted in accordance with FED-STD-H28/20 for the gaging system specified. Thread gages and gaging shall be in accordance with FED-STD-H28/6.

4.4 Grain flow. Grain flow shall be determined by microexamination using a suitable etchant and magnification to reveal the microstructure to meet the requirement of 3.2.

*4.5 Discontinuities. Discontinuities in the threads shall be determined during product discontinuity inspection by magnetic particle method, MIL-STD-1949, liquid penetrant method, MIL-STD-6866, or other surface inspection method specified in the product procurement document. For thread size smaller than 1/4 inch, cutting of the threaded product may be necessary to permit examination. Acceptance or rejection shall be based upon location, position, and depth of discontinuities using visual inspection and microexamination to meet the requirement of 3.3.

5. PACKAGING

Not applicable.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. These cold formed internal screw threads are used where chipless tapping is desired and where other potential benefits of thread forming can be realized, such as increased thread strength, improved machining productivity, and improved pitch diameter control. Materials such as aluminum, brass, copper, low carbon steels, and some corrosion resistant steels are most suitable for cold formed threads. Wrought low-alloy steel with hardness up to 32 HRC has been tested successfully.

*6.2 Design information. Due to minor diameter craters permitted in 3.3.2, effective bearing area on the mating external thread could be reduced. This would reduce the shear area of the external thread.

6.3 Definitions. Thread configuration, nomenclature, definitions and symbols for screw thread configuration are in accordance with FED-STD-H28/1. Nomenclature, Definitions and Letter Symbols for Screw Threads and ANSI/ASME B1.7M and as specified in applicable documents.

*6.3.1 Runout. As applied to screw threads, unless otherwise specified, this term refers to circular runout of major cylinder (for external threads) or minor cylinder (for internal threads) with respect to the pitch cylinder. Circular runout, in accordance with ANSI Y14.5M, controls cumulative variations due to eccentricity and out-of-roundness. The amount of runout is usually expressed in terms of full indicator movement (FIM).

*6.3.2 Circularity (roundness). Circularity or roundness of threads is defined as one-half the variation in pitch diameter of a screw thread around the circumference of the pitch cylinder. This definition differs from that of circularity (roundness) in ANSI Y14.5M which pertains to radial variation. The following roundness terms are used in screw thread gaging practice:

a. Roundness, oval (2 point). One-half the difference between maximum and minimum pitch diameters over 180 degrees of the pitch cylinder circumference.

b. Roundness, multilobe (3 point). One-half the difference between maximum and minimum variations in the positions of one side of an equiangle triangle which envelopes the pitch cylinder and the theoretical positions of that side, as this enveloping triangle is rotated around the circumference of the pitch cylinder.

*6.4 Cross reference of ANSI/ASME standards to FED-STD-H28 documents.

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ANSI/ASME	FED-STD-H28
ASME B1.1	/2
ANSI/ASME B1.2	/6
ANSI/ASME B1.3M	/20
ANSI/ASME B1.7M	/1

*6.5 Subject (key word) listing.

Screw Thread, Class 2B
Screw Thread, Class 3B
Cold Forming Taps
Rolled Threads (Forging)
Chipless Tapping
Cold Formed Threads

TABLE I. Permissible depth of crest discontinuity.

Threads per inch	Basic thread height, h 0.54127P	Permissible depth 20% h [1]	Threads per inch	Basic thread height, h 0.54127P	Permissible depth 20% h [1]
	inch	inch		inch	inch
80	0.0068	0.0014	16	0.0338	0.0068
72	0.0075	0.0015	14	0.0387	0.0077
64	0.0085	0.0017	13	0.0416	0.0083
56	0.0097	0.0019	12	0.0451	0.0090
48	0.0113	0.0023	11 1/2	0.0471	0.0094
44	0.0123	0.0025	11	0.0492	0.0098
40	0.0135	0.0027	10	0.0541	0.0108
36	0.0150	0.0030	9	0.0601	0.0120
32	0.0169	0.0034	8	0.0677	0.0135
28	0.0193	0.0039	7	0.0773	0.0155
27	0.0200	0.0040	6	0.0902	0.0180
24	0.0226	0.0045	5	0.1082	0.0216
20	0.0271	0.0054	4 1/2	0.1203	0.0241
18	0.0301	0.0060	4	0.1353	0.0271

[1] See figure 3.

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6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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